D-SCRIPT: Structure-Aware Deep Learning for Protein Interaction Prediction

Samuel Sledzieski[1*], Rohit Singh[1*], Lenore Cowen[2], Bonnie Berger [1,3]

(1) MIT Computer Science and Artificial Intelligence Lab., Massachusetts Institute of Technology, Cambridge, MA 01239; (2) Department of Computer Science, Tufts University, Medford, MA 02155. (3) Department of Mathematics, MIT, Cambridge, MA 02139.

Existing PPI data, apart from a few model organisms, are sparse

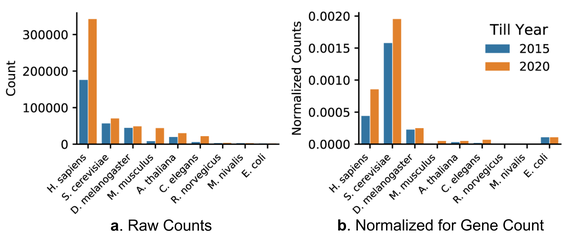

Figure 1: **Counts of PPIs in various organisms detected via direct binding experiments (sourced from BioGRID)**. Shown here are both (**a**) absolute numbers of PPI by species and (**b**) PPI counts normalized by the square of the species gene count. Note the relative lack of PPI data in species other than human and yeast.

Our Contribution

We introduce a new deep learning method, D-SCRIPT, for predicting whether two proteins physically interact (binary label) in the cell, based solely on their amino acid sequences. D-SCRIPT has the following advantages:

- ***Interpretable:*** Lack of interpretability is a shortcoming of many deep learning methods. D-SCRIPT has interpretable intermediate representations.
- ***Generalizable:*** Methods for PPI prediction often fail outside of species where many experimentally derived PPIs are known. D-SCRIPT generalizes better than state-of-the-art methods PIPR [1] and DeepPPI [2] to predict interactions when trained only on interactions from a different species.

[1] M. Chen et al. Multifaceted protein–protein interaction prediction based on Siamese residual RCNN, Bioinformatics, 35(14), 2019, i305–i314. [2] F. Richoux et al. Comparing two deep learning sequence-based models for protein-protein interaction prediction. arXiv. 2019. [3] T. Bepler and B. Berger. Learning protein sequence embeddings using information from structure. International Conference on Learning Representations, 2019.

Methods Overview: D-SCRIPT Architecture

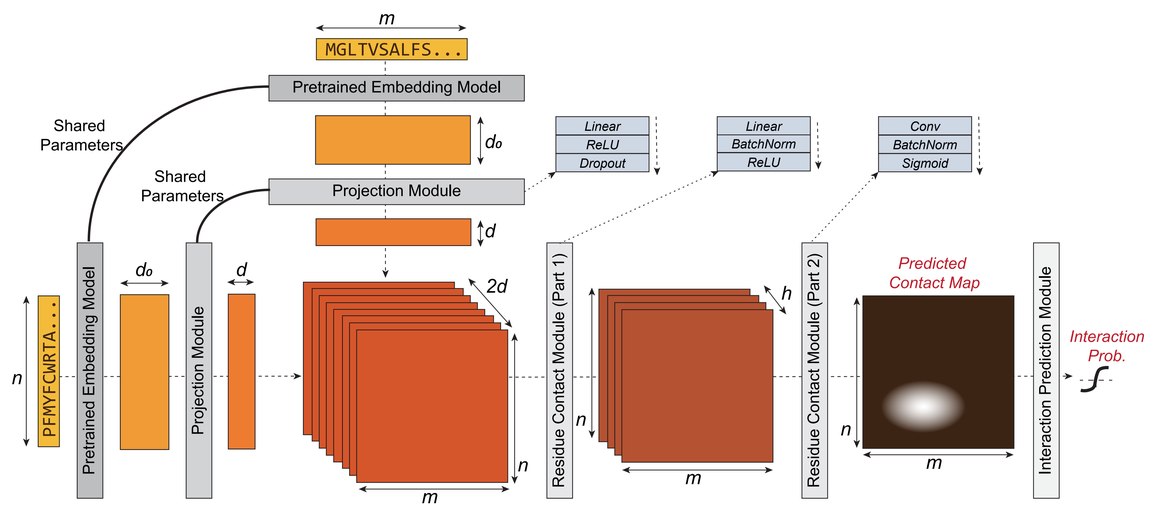

Figure 2: **D-SCRIPT architecture overview.** *Left to right:* A Pre-trained Embedding Mode [3] generates features for each individual protein sequence. The Projection Model reduces them to d dimensions. A sparse contact map is computed through a two-step process which computes a representation for each pair of residues and then incorporates local information with a convolutional filter. A key design goal was that this contact map representation be structurally interpretable. The Interaction Prediction Module then applies a customized max-pooling operation on this map to predict the probability of interaction. The network is trained through error back-propagation, and is supervised using *only* a binary label of protein interaction.

The learned inter-protein contact map is interpretable and informative

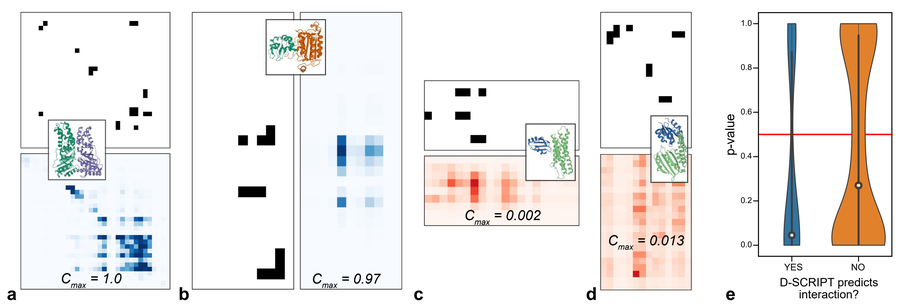

Figure 3: **Inter-protein contact maps of protein structures known to dock together.** Panels **(a,b)** correspond to pairs where D-SCRIPT correctly predicted an interaction, while panels **(c,d)** are cases where it incorrectly predicted no interaction. The black-and-white matrices correspond to the PDB ground truth while the colored matrices correspond to D-SCRIPT's contact map; for the latter, the color scales of **(a,b)** differ from **(c,d)**. C_{max} is the maximum predicted contact between any pair of residues. While the prediction contains large values for positive pairs, its maximum value is very low for negative pairs. Panel **(e)** shows a systematic evaluation of the 2-D Earth mover's distance-based similarity between the predicted and true contact maps. Not only are the correctly-predicted contact maps significantly similar to the ground truth, even when D-SCRIPT incorrectly predicts two proteins don't interact, its contact maps are often still similar to the ground truth.

Results

Table 1: **Evaluation of models trained on human PPIs.** We compared D-SCRIPT with PIPR [1], a state-of-the-art sequence based model for PPI prediction, and a combined model. Interaction data was taken from the STRING database [Version 11] and filtered for high-confidence experimentally determined physical PPIs. Both models were trained using only human protein interactions, and evaluated on interactions from human and 5 model species.

Species	Model	AUPR
M. musculus	PIPR	0.526
	D-SCRIPT	0.580
	PIPR + D-SCRIPT	**0.609**
D. melanogaster	PIPR	0.278
	D-SCRIPT	0.552
	PIPR + D-SCRIPT	**0.562**
C. elegans	PIPR	0.346
	D-SCRIPT	0.548
	PIPR + D-SCRIPT	**0.559**
S. cerevisiae	PIPR	0.230
	D-SCRIPT	0.405
	PIPR + D-SCRIPT	**0.417**
E. coli	PIPR	0.3085
	D-SCRIPT	0.571
	PIPR + D-SCRIPT	**0.588**
H. sapiens (5-fold cross validation)	PIPR	0.835
	D-SCRIPT	0.516
	PIPR + D-SCRIPT	**0.844**

Acknowledgements

We thank Tristan Bepler for helpful discussions and technical assistance. SS, RS, and BB were supported by NIH grant RO1 GM081871. LC was supported by NSF HDR grant 1939263.

- **Download:** `http://dscript.csail.mit.edu`
- **Co-first authors:** S. Sledzieski, R. Singh
- **Co-corresponding authors:** cowen@cs.tufts.edu, bab@mit.edu